

Status Note: High-Rank CBAP/OAP Failure Near $p = 2$

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June 18, 2026

Status. This is a `literature_implied_answer` (partial high-rank subcase). It records a later theorem of de Laat–de la Salle [2] that partially answers an endpoint question raised in de Laat–de la Salle [1].

Source question. In the introduction of [1], after improving the known failure of CBAP and OAP for noncommutative $L^p(L(\Gamma))$ spaces associated with higher-rank lattices from the range

$$[1, 12/11) \cup (12, \infty] \quad \text{to} \quad [1, 10/9) \cup (10, \infty],$$

the authors ask whether this failure can be extended to all $p \neq 2$.

Supporting theorem. The later paper [2], Theorem 1.1, proves that if $n \geq 3$ and $r \geq 2n - 3$, then

$$L^p(L(\mathrm{SL}(r, \mathbb{Z})))$$

does not have CBAP for

$$p \in [1, 2 - 2/n) \cup (2 + 2/(n - 2), \infty].$$

The paper also notes that the same argument gives failure of OAP, and Theorem 1.3 gives the corresponding conclusion for lattices in connected simple real Lie groups of sufficiently high real rank.

Identification. Fix $p \neq 2$. If $p > 2$, choose n so large that $p > 2 + 2/(n - 2)$. If $1 \leq p < 2$, choose n so large that $p < 2 - 2/n$. Then Theorem 1.1 of [2] applies to $\mathrm{SL}(r, \mathbb{Z})$ for every $r \geq 2n - 3$, and Theorem 1.3 applies to sufficiently high-rank real simple Lie groups and their lattices. Thus the source endpoint question has an affirmative answer in the rank-varying high-rank subcase: for every $p \neq 2$, high enough rank lattices have noncommutative L^p -spaces without CBAP and without OAP.

Scope limitations. This does not solve the fixed low-rank problem. In particular, [2] explicitly leaves open whether $L^p(L(\mathrm{SL}(3, \mathbb{Z})))$ has CBAP for some $p \neq 2$. It also does not settle the separate question in [1] about whether the classes $\mathcal{E}_r$ contain all Banach spaces of nontrivial type.

References

- [1] T. de Laat and M. de la Salle, *Strong property (T) for higher rank simple Lie groups*, Proc. Lond. Math. Soc. (3) 111 (2015), no. 4, 936–966; arXiv:1401.3611.
- [2] T. de Laat and M. de la Salle, *Approximation properties for noncommutative L^p -spaces of high rank lattices and nonembeddability of expanders*, J. Reine Angew. Math. 737 (2018), 49–69; arXiv:1403.6415.